INFT CNS MSE I Solution

Total Marks of Question no.		Examiner 1	2/9/24

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	Q.1	
	a)	Classification of Cryptography methods

- Symmetric Key
  - Stream Cipher
    - Substitution Cipher
      - Monoalphabetic
        - Additive
        - Multiplicative
        - Affine
        - Caesar
      - Polyalphabetic
        - Playfair
        - Hill
        - Vigenere
    - Transposition
    - Hybrid
  - Block Cipher
    - DES
    - 2DES
    - 3DES
    - AES
- Asymmetric key
  - RSA
  - elgamal
  - ek

b) P_T = "Mangoes are Sweet and costly"

Key = (11, 9)

Multiplicative Key = 11

Additive Key = 9

Encryption = $[(P_T * K_1) + K_2] \bmod 26$

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i) $C_{T_1} = [(P_1 \times K_1) + K_2] \bmod 26$

$= [(12 \times 11) + 9] \bmod 26$

$= [132 + 9] \bmod 26$

$= [141 \bmod 26$

$= 11$

$C_{T_2} = (0 \times 11 + 9) \bmod 26$

$9 \bmod 26 = 9 = 9$

$C_{T_3} = [(13 \times 11) + 9) \bmod 26$

$= 152 \bmod 26 = 12$

$C_{T_4} = [6 \times 11 + 9] \bmod 26$

$= 75 \bmod 26 = 23$

$C_{T_5} = 14 \times 11 + 9 \bmod 26$

$= 163 \bmod 26 = 7$

$C_{T_6} = (4 \times 91 + 9) \bmod 26$

$= 53 \bmod 26 = 1$

$C_{T_7} = (18 \times 11 + 9) \bmod 26$

$= 207 \bmod 26 = 25$

ARE :-

$C_{T_8} = (0 \times 11 + 9) \bmod 26$

$= 9 \bmod 26 = 9$

$C_{T_9} = (17 \times 11 + 9) \bmod 26$

$= 196 \bmod 26 = 16$

$C_{T_{10}} = (4 \times 11 + 9) \bmod 26$

$= 53 \bmod 26 = 1$

Sweet :-

$C_{T_{11}} = (18 \times 11 + 9) \bmod 26$

$= 207 \bmod 26 = 25$

Total Marks of Question no.		Examiner 1	

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$C_{T12} = (22 \times 11 + 9) \bmod 26$

$= 251 \bmod 26 = 17$

$C_{T13} = (4 \times 11 + 9) \bmod 26$

$= 53 \bmod 26 = 1$

$C_{T14} = (4 \times 11 + 9) \bmod 26$

$= 53 \bmod 26 = 1$

$C_{T15} = (19 \times 11 + 9) \bmod 26$

$= 218 \bmod 26 = 10$

and :-

$C_{T16} = (0 \times 11 + 9) \bmod 26$

$= 9 \bmod 26 = 9$

$C_{T17} = (13 \times 11 + 9) \bmod 26$

$= 152 \bmod 26 = 22$

$C_{T18} = (3 \times 11 + 9) \bmod 26$

$= 43 \bmod 26 = 17$

Costly :-

$C_{T20} = (2 \times 11 + 9) \bmod 26$

$= 31 \bmod 26 = 5$

$C_{T21} = (14 \times 11 + 9) \bmod 26$

$= 163 \bmod 26 = 7$

$C_{T22} = (18 \times 11 + 9) \bmod 26$

$= 207 \bmod 26 = 25$

$C_{T23} = (19 \times 11 + 9) \bmod 26$

$= 218 \bmod 26 = 10$

$C_{T24} = (11 \times 11 + 9) \bmod 26$

$= 130 \bmod 26 = 0$

$C_{T25} = (24 \times 11 + 9) \bmod 26$

$= 273 \bmod 26 = 13$

Total Marks of Question no.		Examiner 1	

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∴ Gr = L.Jwxhbz job zrbbk
Jwq fhzban

c) Multiplicative inverse of 17

$17^{-1} \bmod 26$

∴ $r_1 = 26$
$r_2 = 17$
$t_1 = 0$
$t_2 = 1$

q	r_1	r_2	r	t_1	t_2	t
1	26	17	9	0	1	$(0 - 1\times1) = -1$
1	17	9	8	1	-1	$(1 - 1\times(-1)] = 2$
1	9	8	1	-1	2	$[-1 - (1\times2)] = -3$
8	8	1	0	2	-3	$[2 - (8\times-3)] = 26$
	1	0		-3	26	

∴ MI = 17 = -3 = -3 + 26 = 23

Question no.

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Q.2

a) P: "Security is an important aspect in Communication"

Key : "Cryptography"

Soln

c	r	y	p	t
o	g	a	h	b
d	e	f	i/j	k
m	n	q	s	u
v	w	x		

c	r	y	p	t
o	g	a	h	b
d	e	f	i/j	k
l	m	n	q	s
u	v	w	x	z

5×5

Sc = mk
cu = oc
ri = pe
ty = cp
is = kq
an = fw
im = eq
po = ch
rt = yc
an = fw
ta = yb

sp = qt
ec = dr
tx = pk
nc = lr
om = gl
mu = lv
ni = qf
ca = yo
ti = pk
on = al

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		∴ ℙ = mkoc Pecpkq.fweq chyx.fwyb qldo Pkhlwgl lvq.fyo pkal
	b)	Phishing is a type of cyber attack where attackers attempt to deceive individuals into proving sensitive information such as usernames, passwords, credit card numbers or other personal data. This is typically done by masquerading as a trustworthy entity in electronic communications like emails, text messages or websites. The attacker lures the victim into clicking a malicious link or downloading an infected attachment, leading to compromise of their informatio
		Characteristic of phishing attack:
		1. Deceptive emails & messages
		2. Malicious links or attachments
		3. urgency & fear tactics
		4. Data Harvesting
		Tools used to perform phishing:
		1. Phishing kit
		2. Spear phishing tools
		3. Malware
		4. Social engineering tools
		5. Phishing as a Service
		6. email Spoofing tools

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		7. fake website generators